Lesson 1.1 - Introduction to Statistics

> **Video:** Introduction to Statistics by The Organic Chemistry Tutor **Chapter:** 1 - Introduction to Statistics

This lesson is a fast tour of the basic tools of **descriptive statistics**. It covers summarizing a data set with single numbers (mean, median, mode, range, quartiles), spotting outliers, and drawing the data (box plots, dot plots, stem-and-leaf plots, frequency tables, histograms). Each topic comes back in more detail in later chapters.

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1. Mean, median, mode, and range

Measure	What it is	How to get it
Mean $\bar{x}$	The average	Add all values, divide by how many there are: $\bar{x} = \frac{\sum x}{n}$
Median	The middle value	Sort the data. Odd n: the middle value. Even n: average the two middle values
Mode	The most frequent value	Count repeats. Two modes = **bimodal**; one mode = **unimodal**
Range	The spread from lowest to highest	$\text{range} = \text{highest} - \text{lowest}$

> **Tip:** always sort the data from smallest to largest first. It makes the median, mode and range easy to read off.

Example 1 - an odd number of values

Data (sorted): 7, 7, 10, 14, 15, 23, 32 ($n = 7$)

- Mean: $\bar{x} = \frac{\sum x}{n} = \frac{7+7+10+14+15+23+32}{7} = \frac{108}{7} \approx 15.43$
- Median: cross off values from both ends until one is left → **14**
- Mode: 7 appears twice → **7**
- Range: $32 - 7 = 25$

Example 2 - an even number of values

Data (sorted): 11, 15, 15, 21, 37, 41, 59, 59 ($n = 8$)

- Mean: $\frac{258}{8} = 32.25$
- Median: the two middle values are 21 and 37 → $\frac{21+37}{2} = 29$
- Mode: 15 and 59 both appear twice → **15 and 59** (bimodal)
- Range: $59 - 11 = 48$

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2. Quartiles and the interquartile range (IQR)

Quartiles cut the sorted data into four parts with the same number of values in each:

Symbol	Name	Meaning
Min	Minimum	Smallest value
Q_1	First quartile (25th percentile)	Median of the **lower half**
Q_2	Second quartile (50th percentile)	Median of the **whole** data set
Q_3	Third quartile (75th percentile)	Median of the **upper half**
Max	Maximum	Largest value

$$\text{IQR} = Q_3 - Q_1$$

When n is odd, the overall median is left out of both halves, so the two halves stay the same size.

Outliers: the 1.5 × IQR rule

A value is **not** an outlier if it lies inside the "fences":

$$Q_1 - 1.5\,\text{IQR} \quad \leq \quad x \quad \leq \quad Q_3 + 1.5\,\text{IQR}$$

Anything below the lower fence or above the upper fence is an **outlier**.

Example 3 - quartiles with odd n

Data: 7, 11, 14, 5, 8, 27, 16, 10, 13, 17, 16

Sorted: 5, 7, 8, 10, 11, **13**, 14, 16, 16, 17, 27 ($n = 11$)

- $Q_2 = 13$ (the 6th value)
- Lower half 5, 7, **8**, 10, 11 → $Q_1 = 8$
- Upper half 14, 16, **16**, 17, 27 → $Q_3 = 16$
- $\text{IQR} = Q_3 - Q_1 = 16 - 8 = 8$
- Fences: $Q_1 - 1.5\,\text{IQR} = 8 - 1.5(8) = -4$ and $Q_3 + 1.5\,\text{IQR} = 16 + 1.5(8) = 28$

27 looks far from the rest, but it is inside $[-4, 28]$, so it is **not** an outlier. A value of 29 would have been.

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3. Box-and-whisker plots

A box plot draws the **five-number summary**:

- The **box** runs from Q_1 to Q_3, so its length is the IQR.
- A line inside the box marks the median Q_2.
- The **whiskers** reach out to the smallest and largest values that are **not** outliers.
- Outliers are drawn as separate **points** (`*`) beyond the whiskers.

The four possible shapes: is the min or the max an outlier?

Check the minimum against the lower fence and the maximum against the upper fence. That gives four cases:

(a) Neither is an outlier. The whiskers go all the way to the min and the max.

```
min           Q1      Q2      Q3               max
 |-------------[=======|=======]----------------|
```

(b) Only the max is an outlier. The right whisker stops at the largest value that is **not** an outlier, and the max is drawn as a separate point.

```
min           Q1      Q2      Q3     largest non-outlier     max (outlier)
 |-------------[=======|=======]-----------|                    *
```

(c) Only the min is an outlier. The left whisker stops at the smallest value that is **not** an outlier, and the min is drawn as a separate point.

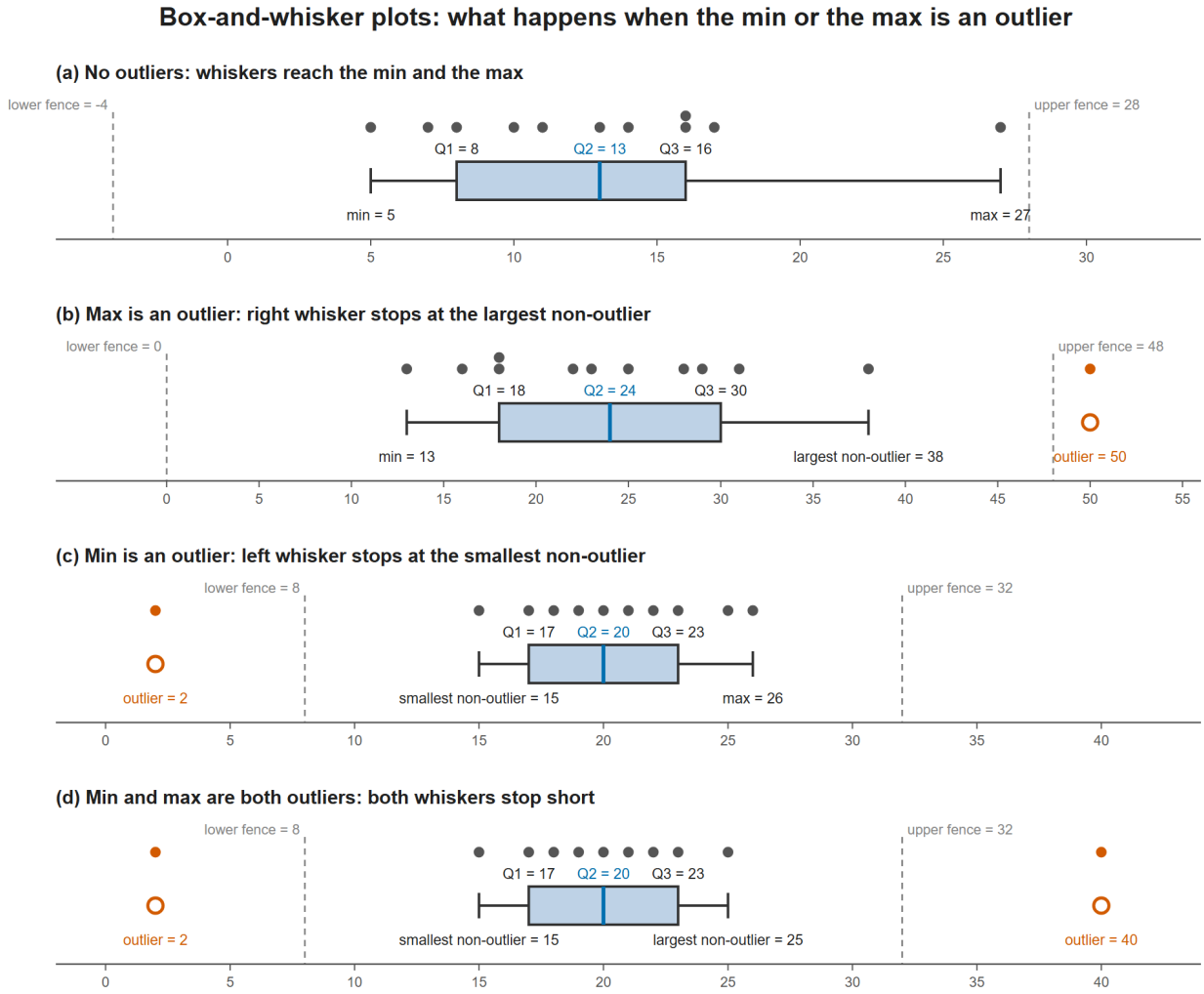
```
min (outlier)     smallest non-outlier   Q1      Q2      Q3          max
 *                        |---------------[=======|=======]-----------|
```

(d) Both are outliers. Both whiskers stop short, and the min and max are both drawn as separate points.

```
min (outlier)   smallest           Q1      Q2      Q3       largest      max (outlier)
 *                 |----------------[=======|=======]----------|            *
```

> The **whisker ends** are therefore not always the min and max. They are always the most extreme values that are **inside** the fences.

The same four cases, drawn to scale from the worked examples below. The dots above each box are the actual data; outliers are orange.



Dots = the data (outliers in orange). Dashed lines = fences Q1 − 1.5×IQR and Q3 + 1.5×IQR. Data from the worked examples in lesson 1.1.

Worked example for case (a) - Example 3 data, no outliers

Using Example 3 from Section 2: min = 5, $Q_1 = 8$, $Q_2 = 13$, $Q_3 = 16$, max = 27, fences $[-4, 28]$. Both 5 and 27 are inside the fences, so the whiskers reach the min and the max.

```
              +---------|-----+
        |-----|         |     |---------------------|
              +---------|-----+
+---------+---------+---------+---------+---------+---------+
0         5         10        15        20        25        30
       min=5   Q1=8      Q2=13 Q3=16               max=27
```

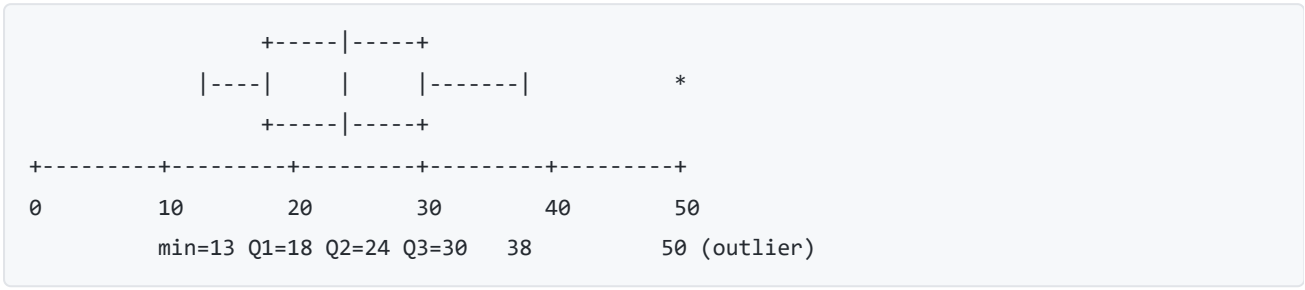
Example 4 - building a box plot ($n = 12$), case (b)

Sorted data: 13, 16, 18 | 18, 22, 23 | 25, 28, 29 | 31, 38, 50

- $Q_2 = \dfrac{23 + 25}{2} = 24$
- Lower half 13, 16, 18, 18, 22, 23 → $Q_1 = \dfrac{18 + 18}{2} = 18$
- Upper half 25, 28, 29, 31, 38, 50 → $Q_3 = \dfrac{29 + 31}{2} = 30$

- $\text{IQR} = Q_3 - Q_1 = 30 - 18 = 12$
- Fences: $Q_1 - 1.5\,\text{IQR} = 18 - 1.5(12) = 0$ and $Q_3 + 1.5\,\text{IQR} = 30 + 1.5(12) = 48$

So **50 is an outlier**. In the plot, the box runs from 18 to 30 with the median at 24. The left whisker goes down to 13, and the right whisker stops at **38**, the largest value that is not an outlier. 50 is drawn as a separate dot.



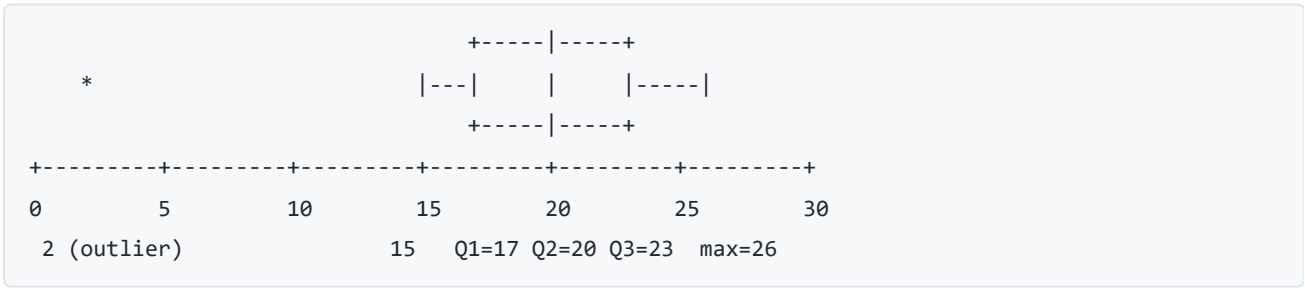
Case (c) - the min is an outlier

These two examples are my own; they are not from the video.

Sorted data ($n = 11$): **2**, 15, 17, 18, 19, **20**, 21, 22, 23, 25, 26

- $Q_2 = 20$
- Lower half 2, 15, **17**, 18, 19 → $Q_1 = 17$
- Upper half 21, 22, **23**, 25, 26 → $Q_3 = 23$
- $\text{IQR} = Q_3 - Q_1 = 23 - 17 = 6$
- Fences: $Q_1 - 1.5\,\text{IQR} = 17 - 1.5(6) = 8$ and $Q_3 + 1.5\,\text{IQR} = 23 + 1.5(6) = 32$

2 is below 8, so the min is an outlier. The left whisker stops at **15**, the smallest value that is not an outlier. The max, 26, is inside the fences, so the right whisker reaches it.



Case (d) - both the min and the max are outliers

Change the largest value in the last data set from 26 to **40**: 2, 15, 17, 18, 19, 20, 21, 22, 23, 25, **40**

- The quartiles don't change: $Q_1 = 17$, $Q_2 = 20$, $Q_3 = 23$, $\text{IQR} = 6$, fences $[8, 32]$.
- **2 < 8** and **40 > 32**, so **both** are outliers.
- The left whisker stops at **15** and the right whisker at **25**, the most extreme values inside the fences. 2 and 40 are drawn as separate points.

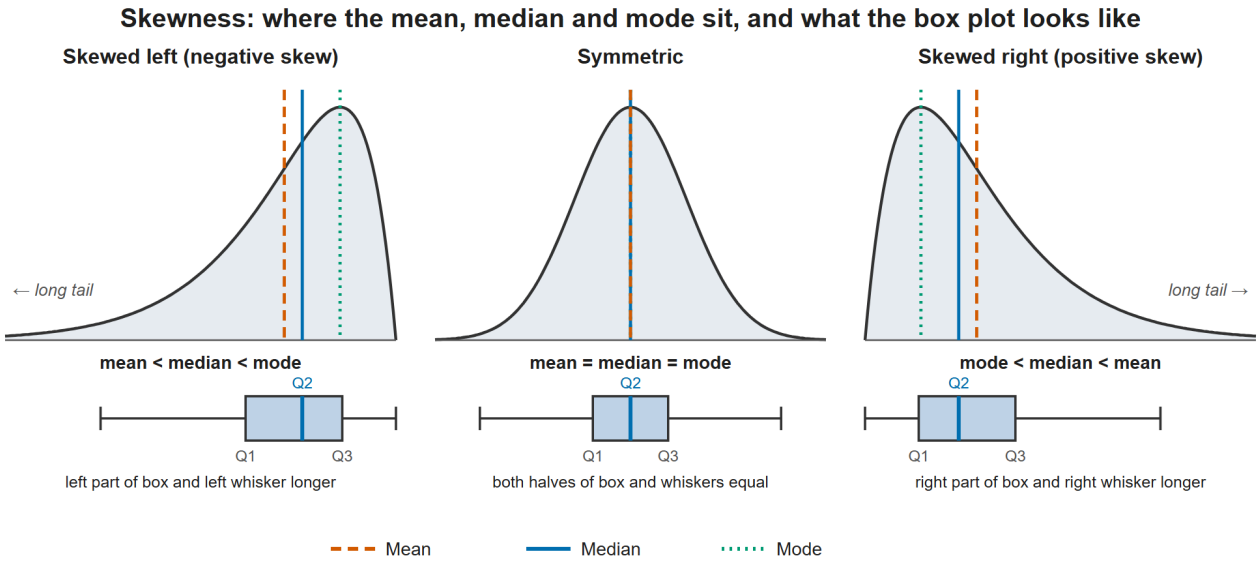
```
                              +-----|-----+
    *                       |---|     |     |---|                         *
                              +-----|-----+
+---------+---------+---------+---------+---------+---------+---------+---------+
0         5         10        15        20        25        30        35        40
 2 (outlier)               15   Q1=17 Q2=20 Q3=23 25                  40 (outlier)
```

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4. Skewness

Shape	Tail points to	Mean vs median	Box plot clue
Symmetric	No long tail	mean ≈ median	Median in the middle of the box; whiskers equal
Skewed right (positive skew)	Right	mean > median	Right part of the box ($Q_3 - Q_2$) and/or right whisker longer
Skewed left (negative skew)	Left	mean < median	Left part of the box ($Q_2 - Q_1$) and/or left whisker longer

> **Memory aid:** the mean gets pulled toward the tail. A right tail drags the mean to the right of the median.



Curves: Gamma(2, 1), a normal curve, and the mirror image of Gamma(2, 1). Whiskers drawn to the 1.5 × IQR fences.

In each panel the box plot is on the same scale as the curve above it. In the skewed panels, the longer side of the box and the longer whisker point toward the tail.

The figures in this note were drawn for these notes. `figures/make_figures.ipynb` recreates them with matplotlib, and SVG versions (skewness, box plots, histogram) are in the same folder.

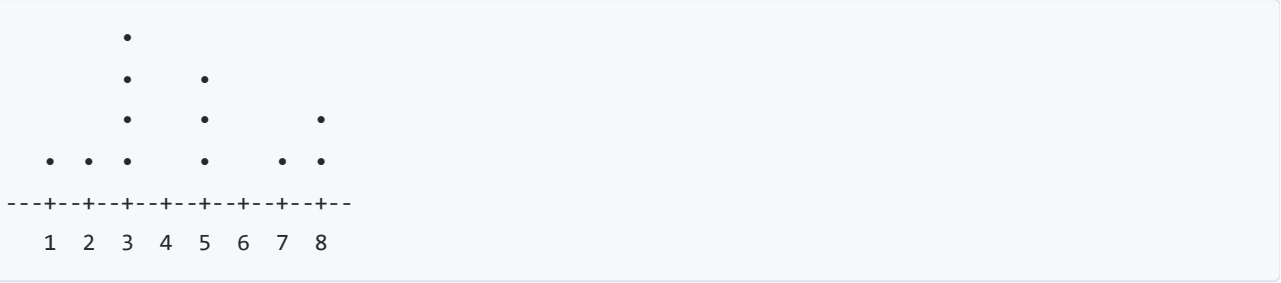
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5. Dot plots

Draw a number line and put one dot above a value each time it appears. The tallest column of dots is the **mode**.

Example 5. Data: 5, 8, 3, 7, 1, 5, 3, 2, 3, 3, 8, 5

Counts: 1 → 1, 2 → 1, 3 → 4, 5 → 3, 7 → 1, 8 → 2



The tallest column is above 3, so the **mode is 3**.

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6. Stem-and-leaf plots

Split each number into a **stem** (the leading digit or digits) and a **leaf** (the last digit). Write the leaves in order, and keep a stem even if it has no leaves. Always add a **key**.

Example 6. Data: 4, 9, 13, 13, 17, 21, 36, 38, 38, 56

Stem	Leaf
0	4 9
1	3 3 7
2	1
3	6 8 8
4	
5	6

Key: 2 | 1 = 21

Example 7 - three-digit numbers. Data: 78, 85, 89, 92, 106, 107, 119

Stem	Leaf
7	8
8	5 9
9	2
10	6 7
11	9

Key: 10 | 6 = 106

Example 8 - decimals. Data: 1.2, 1.6, 1.8, 2.1, 2.3, 2.3, 2.5

Stem	Leaf
1	2 6 8
2	1 3 3 5

Key: 1 | 6 = 1.6

---

7. Frequency tables and the mean

A **frequency table** lists each value and how many times it occurs. Multiplying each value by its frequency lets you find the mean without writing every number out:

$$\bar{x} = \frac{\sum (x \cdot f)}{\sum f}$$

Example 9. Data: 5, 9, 8, 7, 8, 12, 9, 8, 10, 8, 9, 7

Value x	Frequency f	$x \cdot f$
5	1	5
7	2	14
8	4	32
9	3	27
10	1	10
12	1	12
Total	**12**	**100**

$$\bar{x} = \frac{\sum xf}{\sum f} = \frac{100}{12} \approx 8.33$$

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8. Histograms

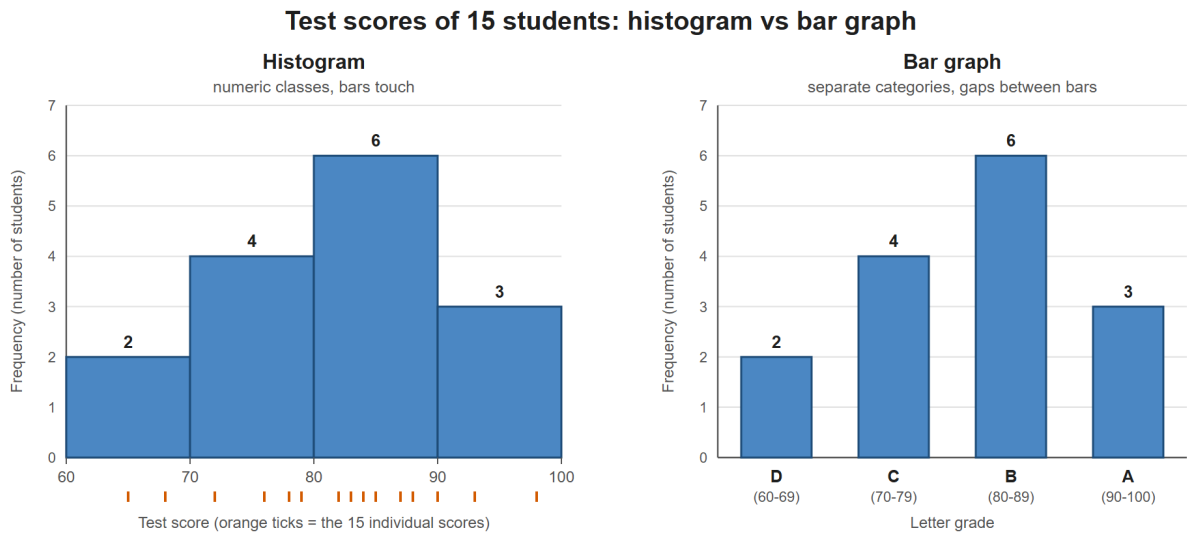
A **histogram** looks like a bar graph, but its bars **touch**. Each bar covers a range of values (a *class*), and its height is the frequency. A bar graph has gaps between its bars and is used for separate categories.

Example 10. Test scores: 65, 72, 93, 68, 76, 98, 84, 85, 79, 88, 90, 82, 83, 87, 78

First group the scores into classes:

Class (grade)	Scores	Frequency
60-69 (D)	65, 68	2
70-79 (C)	72, 76, 79, 78	4
80-89 (B)	84, 85, 88, 82, 83, 87	6
90-100 (A)	93, 98, 90	3
Total		**15**

Then draw adjacent bars of heights 2, 4, 6 and 3 over the classes 60-70, 70-80, 80-90 and 90-100.



Scores: 65, 68 | 72, 76, 78, 79 | 82, 83, 84, 85, 87, 88 | 90, 93, 98. Classes 60-69, 70-79, 80-89, 90-100.

On the left, the histogram's bars sit on a **number line** and touch, because the classes run straight into each other (70 is where 60-69 ends and 70-79 starts). The orange ticks under the axis are the 15 individual scores, so you can see which bar each one falls into. On the right, the same counts are drawn as a **bar graph** of the letter grades D, C, B and A, which are separate categories, so the bars have gaps.

The tallest bar is 80-89, so most students got a B. The shape has a longer tail on the low side (60s), so it leans slightly left.

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9. Relative and cumulative relative frequency, and percentiles

- **Relative frequency** = $\frac{f}{n}$, the fraction of the data at that value. The column adds up to 1.
- **Cumulative relative frequency** is the running total of the relative frequencies. It ends at 1.

Example 11. Data ($n = 20$): 2, 3, 5, 3, 6, 8, 7, 8, 3, 3, 5, 3, 7, 3, 8, 5, 2, 7, 8, 3

Value	Frequency	Relative frequency	Cumulative relative frequency
2	2	0.10	0.10
3	7	0.35	0.45
5	3	0.15	0.60
6	1	0.05	0.65
7	3	0.15	0.80
8	4	0.20	1.00
Total	**20**	**1.00**	

Reading percentiles from the table

The method used in the video:

- If the percentile lands **exactly** on a cumulative value, average that value and the next one.
  - 60th percentile: the cumulative column hits 0.60 at value 5, so $\frac{5+6}{2} = 5.5$.

    - 80th percentile: it hits 0.80 at value 7, so $\frac{7+8}{2} = 7.5$.
- If the percentile falls **between** two cumulative values, take the value of the higher row.
    - 20th percentile: between 0.10 and 0.45, so it is **3**.
    - 75th percentile: between 0.65 and 0.80, so it is **7**.

You can check this by writing all 20 sorted values in a row and marking every 10%, which falls every 2 values:

```
2 2 | 3 3 | 3 3 | 3 3 | 3 5 | 5 5 | 6 7 | 7 7 | 8 8 | 8 8
   10%   20%   30%   40%   50%   60%   70%   80%   90%  100%
```

The 60% mark sits between a 5 and a 6 (→ 5.5). The 20% mark sits between two 3s (→ 3).

> Textbooks and software use several slightly different percentile rules, so small differences between methods are normal.

Key takeaways

- **Sort first**, then find the median, quartiles, mode and range.
- **Mean** is affected by extreme values. The **median** and **IQR** are resistant to them.
- **Outlier fences:** $Q_1 - 1.5\,\mathrm{IQR}$ and $Q_3 + 1.5\,\mathrm{IQR}$.
- **Box plot:** box = IQR, line = median, whiskers = most extreme non-outliers, dots = outliers.
- **Skew:** the mean is pulled toward the tail.
- **Histograms** have touching bars over numeric classes; **bar graphs** have gaps between categories.
- **Relative frequency** = f/n; the **cumulative** column ends at 1 and is used to read percentiles.

Lesson 1.2 - Descriptive Statistics vs Inferential Statistics

> **Video:** Descriptive Statistics vs Inferential Statistics by The Organic Chemistry Tutor **Chapter:** 1 - Introduction to Statistics

What is statistics?

Statistics is the science of **collecting, analyzing, interpreting and presenting data**. It has two main branches:

	Descriptive statistics	Inferential statistics
Goal	**Organize and summarize** the data you have	**Draw conclusions about a population** from a sample
Tools	Graphs, tables, summary numbers	Probability, estimates, confidence intervals, hypothesis tests
Typical statement	"20 of the 100 people surveyed said yes."	"About 20% of the whole city would say yes, ± 2%."

You usually learn descriptive statistics first, because inference is built on top of it.

Descriptive statistics

Descriptive statistics describes data in two ways.

1. With graphs and tables

- **Bar graph:** separate bars, one per category.
- **Histogram:** like a bar graph, but the bars **touch** because they cover consecutive numeric ranges.
- **Pie chart:** shows how a whole is split into percentages.
- **Line graph:** shows change, often over time.
- **Frequency table / frequency distribution table:** counts how often each value or class occurs.
- **Shape of the data:** symmetric, skewed right, or skewed left.

2. With numbers

- **Measures of central tendency:**
  - **Mean** (average) = sum of values ÷ number of values.
    - *Sample mean* $\bar{x}$: averages the values in a sample.
    - *Population mean* μ: averages every value in the population.
  - **Median**: the middle value of the sorted data.
  - **Mode**: the most frequent value.
- **Measures of spread:** range, variance, standard deviation.

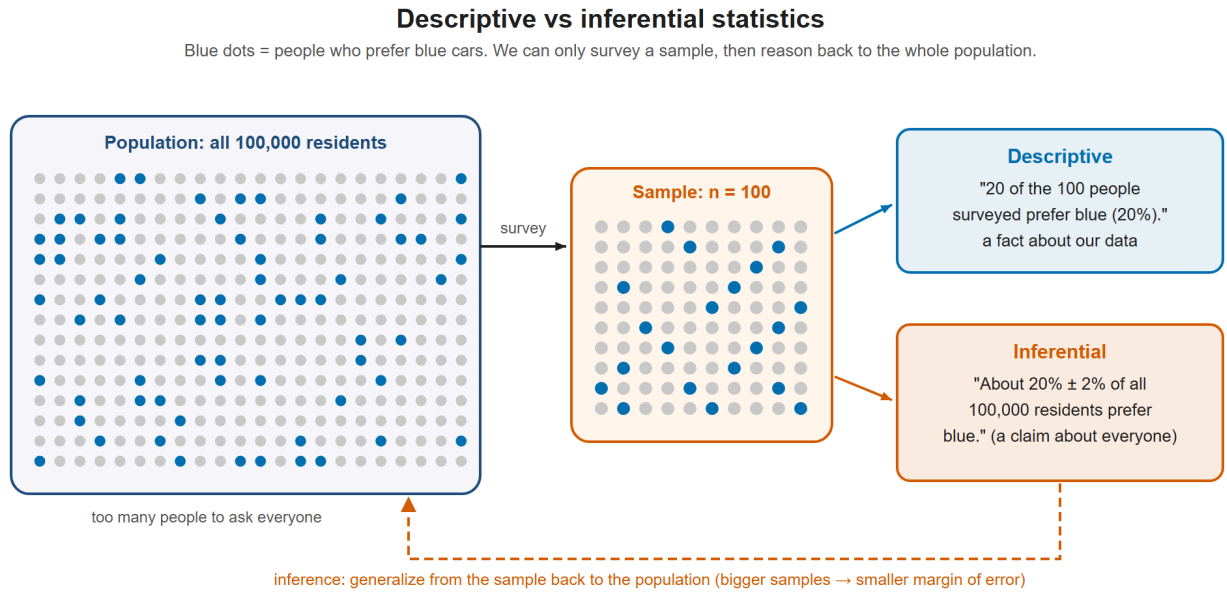
Inferential statistics

Inferential statistics uses **sample data** to make an **inference**, or draw a conclusion, about a **population**. It also uses probability to say **how confident** we can be in that conclusion.

Example - who likes blue cars?

City XYZ has **100,000** residents. We want to know what share of them prefer blue cars.

1. Asking all 100,000 people is not practical, so we survey a **sample of 100** people.
2. **20 out of 100** say they prefer blue cars, so **20%** of the sample. → This is a fact about the data we collected: **descriptive statistics**.
3. We then say: "About 20% of *all* residents prefer blue cars, within a 2% margin of error." → This is a conclusion about people we did not ask: **inferential statistics**.
4. A **confidence interval** is also inferential: "We are 95% confident that 20% ± 2% of residents prefer blue cars."



Describing the 100 people we surveyed is descriptive statistics. Using them to say something about all 100,000 residents is inferential statistics.

A bigger sample gives a better estimate

Suppose we survey **1,000** people instead and **21%** say yes. The margin of error shrinks to about **±1%**. A larger sample gives a more precise picture of the population.

Sample size	Sample result	Margin of error
100	20%	± 2% (as stated in the video)
1,000	21%	± 1% (as stated in the video)

> **Note:** the margins of error here are illustrative numbers from the video. Chapter 9 shows how to actually calculate a margin of error for a proportion. With the standard formula, a sample of 100 at 95% confidence gives a margin of about ±8%. The idea is still the same: **more data → smaller margin of error**.

Key takeaways

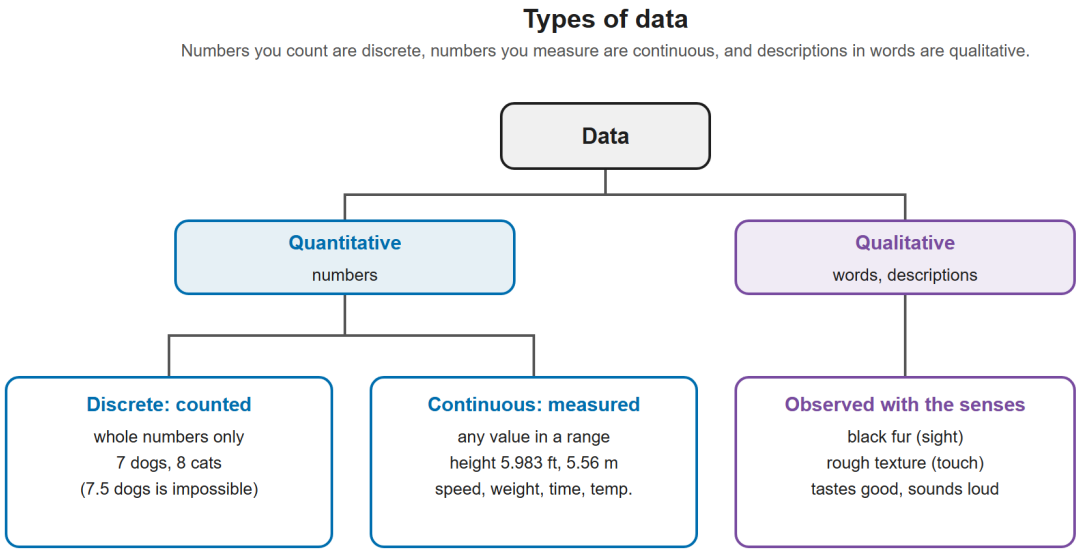
- **Descriptive** = summarize what you observed (graphs, tables, mean, median, mode, range, SD).
- **Inferential** = generalize from a **sample** to the **population**, with a stated level of confidence.
- Reporting "20 of 100 people said yes" is descriptive. Saying "20% of the city agrees" is inferential.
- Bigger samples give more precise inferences (smaller margin of error).

Lesson 1.3 - Quantitative vs Qualitative Data

> **Video:** Qualitative and Quantitative by The Organic Chemistry Tutor **Chapter:** 1 - Introduction to Statistics

Two types of data

```
                    Data
          ┌─────────┴─────────┐
   Quantitative              Qualitative
(numbers / numerical)    (words / descriptive)
     ┌────┴────┐
 Discrete   Continuous
(counted)   (measured)
```



The same tree as above, with an example in each box.

Quantitative data - numbers

Quantitative data is **numerical**, such as 5, 8 or 12. It comes in two forms.

Discrete - things you **count**

- Only whole-number values (0, 1, 2, 3, ...); nothing in between.
- Example: **8 cats**. You cannot have 8.5 cats or 2.37 cats.

Continuous - things you **measure**

- Any value in a range, including decimals.
- Example: a distance of 5.56 m or 7.283 m. It does not have to be a whole number.

- Other examples: **distance, speed, weight, height, time, temperature**.

> **Rule of thumb:** counted → discrete; measured → continuous. Both are quantitative.

Qualitative data - words

Qualitative data is **descriptive**. It comes from **observation** and is described with **words, not numbers**. It often involves the five senses:

Sense	Example
Sight	Color (the fur is black)
Touch	Texture (soft, hard, rough)
Taste	The food tastes good / bad
Hearing	The volume is low / high
Smell	It smells good / bad / of nothing

Quick quiz

Statement	Quantitative or qualitative?	Discrete or continuous?	Why
There are **7 dogs**	Quantitative	**Discrete**	You count dogs; 7.5 dogs is impossible
The cat's fur is **black**	**Qualitative**	n/a	Color, described with a word
Michael is **6 feet tall**	Quantitative	**Continuous**	Height is measured; it could be 5.983 ft
The cat's fur is **rough**	**Qualitative**	n/a	Texture, described with a word

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Key takeaways

- **Quantitative** = numbers. **Qualitative** = words or descriptions.
- Quantitative data is **discrete** (counted, whole numbers) or **continuous** (measured, any value in a range).
- Qualitative data usually describes what you see, feel, taste, hear or smell.

Lesson 1.4 - Statistic vs Parameter and Population vs Sample

> **Video:** Statistic vs Parameter & Population vs Sample by The Organic Chemistry Tutor **Chapter:** 1 - Introduction to Statistics

Population vs sample

- **Population:** *everyone or everything* in the group being studied.
- **Sample:** a **subset**, a smaller part of the population that is actually measured.

Statistic vs parameter

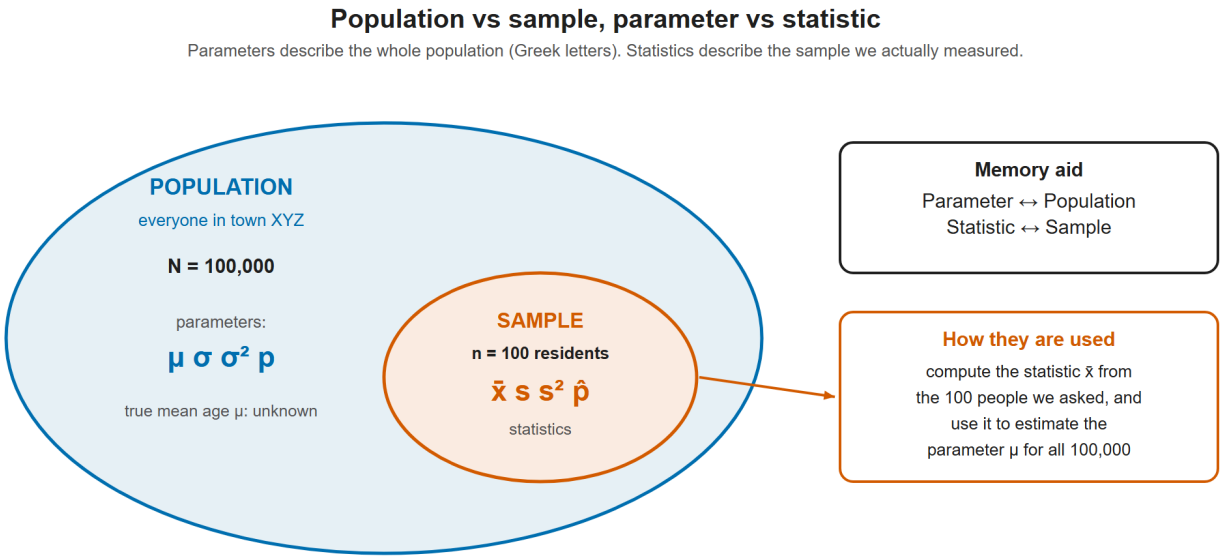
- **Parameter:** a number that describes the **population**.
- **Statistic:** a number that describes a **sample**.

> **Memory aid:** **P**arameter ↔ **P**opulation, **S**tatistic ↔ **S**ample.

Example - average age in town XYZ

Town XYZ has **100,000** residents, and we want their average age.

- Asking all 100,000 people is a lot of work, so we ask **100** residents and average their ages.
- The average age of those 100 people is the **sample mean** $\bar{x}$, which is a **statistic**.
- The true average age of all 100,000 residents is the **population mean** μ, which is a **parameter**.
- Population size = 100,000; sample size = 100.



The sample is part of the population. Numbers computed from the sample ($\bar{x}$, s, $\hat{p}$) are statistics, and we use them to estimate the population's parameters (μ, σ, p).

Symbols

Quantity	Sample (statistic)	Population (parameter)
Mean	$\bar{x}$ ("x-bar")	μ ("mu")
Standard deviation	s	σ ("sigma")
Variance	s^2	σ^2
Proportion	$\hat{p}$ ("p-hat")	p
Size	n (lowercase)	N (uppercase)

Roman letters ($\bar{x}$, s, $\hat{p}$, n) are usually used for samples. Greek letters (μ, σ) and plain p, N are used for populations.

Quick quiz - statistic or parameter?

A parameter describes **everyone** in the group. A statistic describes **only some** of them.

#	Statement	Answer	Symbol
1	The average weight of **all** males in the United States	**Parameter**	μ
2	The average height of **100 cats** in California (out of millions)	**Statistic**	$\bar{x}$
3	The average test score of **20 students** in a class of 500	**Statistic**	$\bar{x}$
4	The average test score of **all** students in the class	**Parameter**	μ

Key takeaways

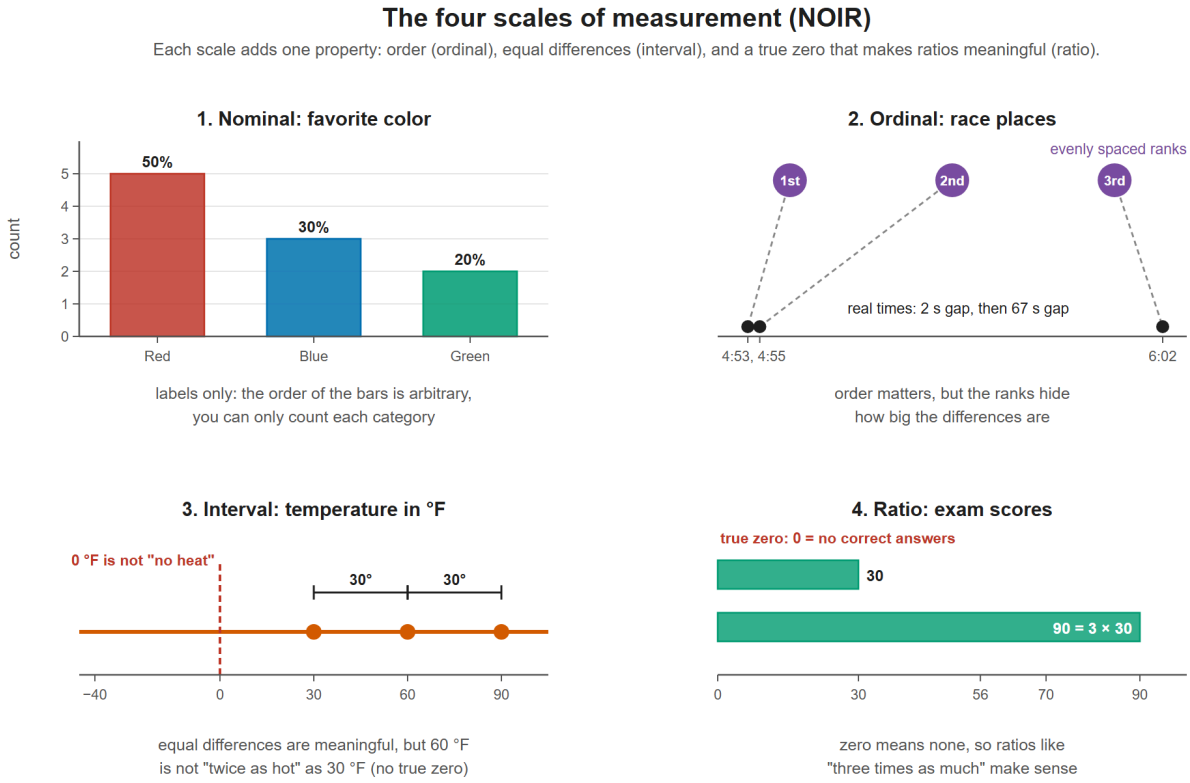
- **Population** = the whole group. **Sample** = the part you actually observe.
- **Parameter** describes a population. **Statistic** describes a sample.
- In practice we usually compute statistics ($\bar{x}$, s, $\hat{p}$) to **estimate** unknown parameters (μ, σ, p).

Lesson 1.5 - Scales of Measurement

> **Video:** Scales of Measurement - Nominal, Ordinal, Interval, & Ratio Scale Data by The Organic Chemistry Tutor **Chapter:** 1 - Introduction to Statistics

Data can be measured on four **scales of measurement**. Each scale adds one more property than the one before it:

Nominal → Ordinal → Interval → Ratio



One example per scale, from the lesson. Each panel shows the property that scale adds: nothing beyond labels (nominal), order (ordinal), equal differences (interval), and a true zero (ratio).

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1. Nominal scale - names and labels

- **Qualitative** and **categorical**: names, colors, labels.
- **Order does not matter.** You might code red = 1, blue = 2, green = 3, but the numbers are arbitrary. Red is not "less than" blue.
- You cannot do arithmetic with the categories themselves, but you **can** count how many fall in each category.

Example. 10 people are asked their favorite color: 5 say red, 3 blue, 2 green.

Color	Count	Percentage
Red	5	50%
Blue	3	30%
Green	2	20%

The percentages are calculated from the **counts**, not from the colors.

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2. Ordinal scale - ranked order

- **Order matters**, so values can be ranked.
- **Differences cannot be measured**: the gap between 1st and 2nd is not necessarily the same as between 2nd and 3rd.

Example 1 - a race.

Place	Finishing time
1st	4 min 53 s
2nd	4 min 55 s
3rd	6 min 02 s

1st and 2nd are only 2 seconds apart, but 2nd and 3rd are more than a minute apart. The places alone (1, 2, 3) hide this. If you only know the ranking, you cannot measure the differences.

Example 2 - a rating scale. Excellent (1), Good (2), Satisfactory (3), Bad (4). We know excellent beats good and good beats satisfactory. We cannot say whether the step from excellent to good is the same size as the step from good to satisfactory.

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3. Interval scale - equal differences, no true zero

- **Order matters.**
- **Differences can be measured**, and they have a consistent meaning.
- **Ratios are not meaningful.**
- **No true zero:** zero does not mean "none of the quantity".

Example - temperature in °F or °C.

- 30 °F, 60 °F, 90 °F are in order.
- 60 − 30 = 30 °F and 90 − 60 = 30 °F are equal, meaningful differences.
- But 60 °F is **not** "twice as hot" as 30 °F, even though 60 / 30 = 2.
- 0 °F is not "no heat": temperatures can be negative (−20 °F, −40 °C).

> Temperature in **kelvin** is different. It has a true zero (absolute zero), so kelvin is a **ratio** scale.

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4. Ratio scale - everything, including a true zero

- **Order matters.**
- **Differences can be measured.**
- **Ratios are meaningful.**
- **True zero exists:** zero means none, and negative values are impossible.

Example - exam scores. Scores 70, 30, 56, 82, 90 sort to 30, 56, 70, 82, 90.

- Differences: 56 − 30 = 26, 70 − 56 = 14.
- Ratio: 90 / 30 = 3, so the top score is three times the lowest.
- A score of 0 means no correct answers (or no test taken), and a negative score is impossible.

Other ratio examples: height, weight, distance, time, money, counts.

Summary table

Property	Nominal	Ordinal	Interval	Ratio
Can be labeled / categorized	✅	✅	✅	✅
Meaningful order	❌	✅	✅	✅
Measurable differences	❌	❌	✅	✅
Meaningful ratios	❌	❌	❌	✅
True zero starting point	❌	❌	❌	✅

> **Memory aid - "NOIR":** **N**ominal, **O**rdinal, **I**nterval, **R**atio, from the least information to the most.